

Cause of Death among People Experiencing Homelessness in Toronto*

Acute drug toxicity causes three in four deaths before age 40 but fewer than one in four from age 60

Graham Sayle

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From 2022 to 2024, Toronto Public Health published the records of 825 deaths of people experiencing homelessness. In this paper these counts are used to describe what they died of and at what age. Of these deaths, 457 were caused by acute drug toxicity, which accounts for 77% of deaths of people aged 20 to 39 but only 24% of those aged 60 and over. This is important to highlight because the two groups need different responses: drug policy for the younger population, and medical and long-term care for the older.

1 Introduction

Toronto's unhoused population dies far earlier than the housed population. A cohort study of Toronto residents found 2.7 deaths per 100 person-years against 0.7 among matched housed residents, with those who died on average seventeen years younger (Richard et al. 2024). Drug poisoning accounts for a growing share of those deaths: among all opioid-related overdose deaths in Ontario, the proportion involving someone who was unhoused rose from 7.2% in late 2017 to 16.8% by mid-2021 (Booth et al. 2024). These studies show that unhoused people die younger, and that they account for a rising share of Ontario's opioid deaths, but neither sets out how cause of death varies with age.

This paper uses the counts of deaths by cause, age group and gender that Toronto Public Health published for 2022 to 2024. As Section 2.6 shows, acute drug toxicity accounts for most recorded deaths and its share falls sharply with age, while cardiovascular disease and other causes follow the reverse pattern. A population dying of disease in later life needs medical and

*Code and data supporting this analysis are available at: https://github.com/GrahamSayle/Homelessness_Deaths

long-term care; one dying of drug poisoning in its twenties and thirties needs a drug policy response, and loses far more years of life to each death.

The remainder of this paper is structured as follows. Section 2 introduces the dataset and explains how a death in the city becomes a record, covering the completeness of recent records in Section 2.3, cause of death in Section 2.4, age and gender in Section 2.5, and the relationship between the two in Section 2.6. Section A records the cleaning steps and the tests applied to the data.

2 Data

2.1 Overview

The data come from the Deaths of People Experiencing Homelessness dataset (Toronto Public Health 2025a), published by Toronto Public Health (TPH) through the City of Toronto Open Data Portal. We obtain and analyse them in R (R Core Team 2024), drawing on the tidyverse collection of packages (Wickham et al. 2019), opendatatoronto (Gelfand 2022) for programmatic access to the portal, here (Müller 2020) for file paths, ggplot2 (Wickham 2016) for figures, tinytable (Arel-Bundock 2025) for tables, and knitr (Xie 2025) for rendering. Following Alexander (2023), we simulate the dataset we expect before acquiring it, which forces us to specify its structure and the relationships we anticipate in advance of seeing them; that simulation, and the tests applied to both the simulated and the actual data, are written in Python (Python Software Foundation 2025) using polars (Vink and Polars Contributors 2025) and numpy (Harris et al. 2020), with the environment managed by uv (Astral Software Inc. 2025).

TPH leads the collection, analysis, and reporting of deaths of people experiencing homelessness in Toronto, drawing on records contributed by Toronto Shelter and Support Services, community agencies, the Toronto Homeless Memorial, and the Office of the Chief Coroner (OCC) for Ontario.

The portal publishes four related resources for this dataset, breaking deaths down by month, by demographic characteristics, by setting of death, and by cause of death. The last of these is used, which is the only resource that crosses cause of death with both age group and gender. The data are published as aggregated counts rather than as individual records: each row reports the number of deaths for one combination of year, cause of death, age group, and gender. The extract contains 125 such rows covering the years 2022 through 2024, and records 825 deaths in total — 320 in 2022, 298 in 2023, and 207 in 2024. Cause of death is recorded in eight categories: acute drug toxicity, cardiovascular disease, homicide, other diseases, suicide, unintentional injury, and two categories for deaths whose cause has not been established, recorded as “pending” and “unknown.”

The portal describes a rule of suppression as redacting small counts to maintain privacy and confidentiality, however this does not appear to be the case. There are 37 cells with a count of 1, but only 7 suppressed cells. However, every cell with age <20 is suppressed, so perhaps the suppression is to preserve the privacy and confidentiality of the youngest population specifically.

An alternative to the resource used would have been the demographics or setting resources published alongside it. The cause of death resource was preferred because our interest is in how cause varies across groups, and neither of the alternatives provided cause.

Table 1 shows a sample of six rows from the cleaned data, chosen to include one suppressed cell alongside typical ones.

Table 1: Sample rows from the cleaned dataset

Year	Cause of death	Age group	Gender	Count	Suppressed
2022	Acute Drug Toxicity	20-39	Female	23	FALSE
2024	Pending	60+	Female	1	FALSE
2024	Cardiovascular Disease	60+	Male	4	FALSE
2023	Homicide	20-39	Female	2	FALSE
2024	Suicide	20-39	Male	2	FALSE
2022	Other Diseases	<20	Male	NA	TRUE

Each row is a count of deaths for one combination of year, cause of death, age group, and gender, rather than a record of an individual death. **Year** takes one of three values, 2022 to 2024. **Cause of death** takes one of eight values: Acute Drug Toxicity, Cardiovascular Disease, Homicide, Other Diseases, Pending, Suicide, Unintentional Injury, and Unknown, where Pending marks a death still under investigation and Unknown marks one where a cause was never established. **Age group** takes one of five values: <20, 20-39, 40-59, 60+, and Unknown. **Gender** takes one of three values: Female, Male, and Unknown. **Count** is the number of deaths matching that combination, and is a whole number when reported; **Suppressed** is TRUE for cells where the portal withholds the count to protect privacy, in which case **Count** is recorded as missing rather than zero.

2.2 Measurement

Toronto Public Health uses a broad definition of homelessness: a person without stable, permanent, appropriate housing, or the immediate prospect, means and ability to acquire it, including someone staying temporarily with friends or family after a long period without permanent housing (Toronto Public Health 2025b).

A death becomes an entry through one of four channels (Toronto Public Health 2025b). Community agencies submit reports through a secure online form. The City’s Shelter and Support Services division reports the deaths of shelter residents. The Toronto Homeless Memorial, a community-run record, supplies monthly summaries. Since 2024, the Office of the Chief Coroner of Ontario has shared death investigation data. Each channel reaches a different part of the population, and the portal notes that the data reflect only deaths reported to TPH and may underestimate the true number. The counts are best read as deaths known to a set of institutions, not as all deaths, and are thus a lower bound on the true numbers.

The addition of coroner data is the clearest illustration of how much the measurement instrument shapes the measurement. Toronto Public Health notes that adding this channel in 2024 increased the number of reported deaths from 2022 onward (Toronto Public Health 2025b).

Finally, the counts carry no denominator. The file records deaths, not the population at risk, so it supports statements about the composition of deaths and not about rates.

2.3 Year

Figure 1 shows the number of deaths recorded in each year. Recorded deaths fall from 320 in 2022 to 207 in 2024. Whether this reflects fewer deaths or fewer being reported cannot be established from these data.

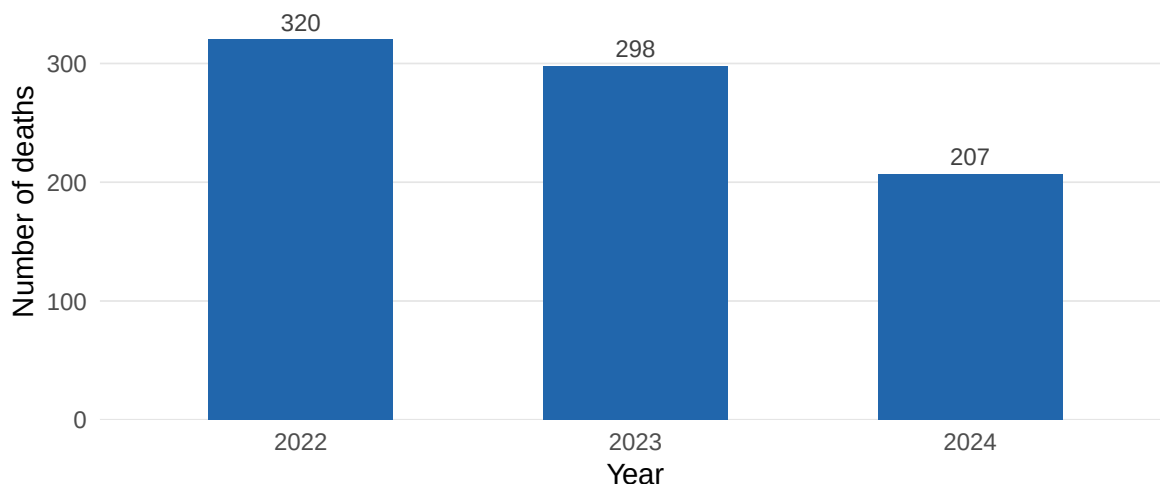


Figure 1: Deaths of people experiencing homelessness in Toronto by year, 2022 to 2024. Counts exclude suppressed cells.

2.4 Cause of death

The number of deaths recorded against each cause across the whole period are shown in Figure 2. Acute drug toxicity accounts for 457 of the 825 deaths, or 55%, which is more than the other seven categories combined. No other single cause reaches 110. The second largest category is not a cause at all but the absence of one: 104 deaths are recorded as unknown.

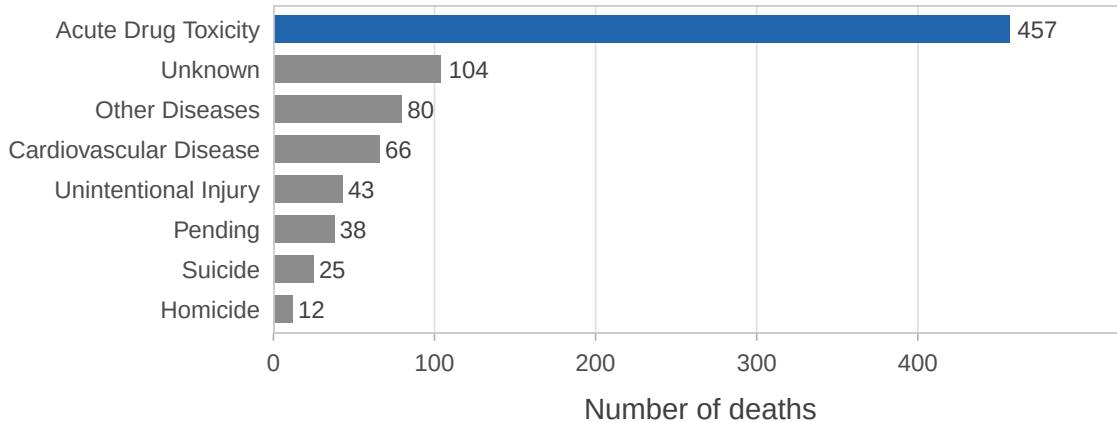


Figure 2: Deaths of people experiencing homelessness in Toronto by cause of death, 2022 to 2024. Cells with counts too small to publish are suppressed and excluded.

Table 2 shows that the same general trend is observed across the three years, with Acute Drug Toxicity being the dominant cause with Cardiovascular Disease and Other Diseases at much lower counts.

Table 2: Deaths of people experiencing homelessness in Toronto by cause of death and year, 2022 to 2024

Cause of death	2022	2023	2024
Acute Drug Toxicity	171	174	112
Unknown	44	42	18
Other Diseases	35	25	20
Cardiovascular Disease	35	19	12
Unintentional Injury	17	17	9
Pending	4	6	28
Suicide	12	9	4
Homicide	2	6	4
Total	320	298	207

2.5 Age and gender

Figure 3 shows every recorded death by age group and gender. Ages below 60 dominate the count, with 40-59 accounting for 43% of the deaths and 20-39 accounting for 31%. 60 and over only had 23%.

Four in five deaths were men, and an imbalance is seen in every age group. The percentage of deaths that were male per age group rises with age, being 70% for ages 20-39, 83% for ages 40-59, and 85% for ages 60 and over. Given that the portal gives no information on the total counts of unhoused people, no inference can be made on the rates of death of the population.

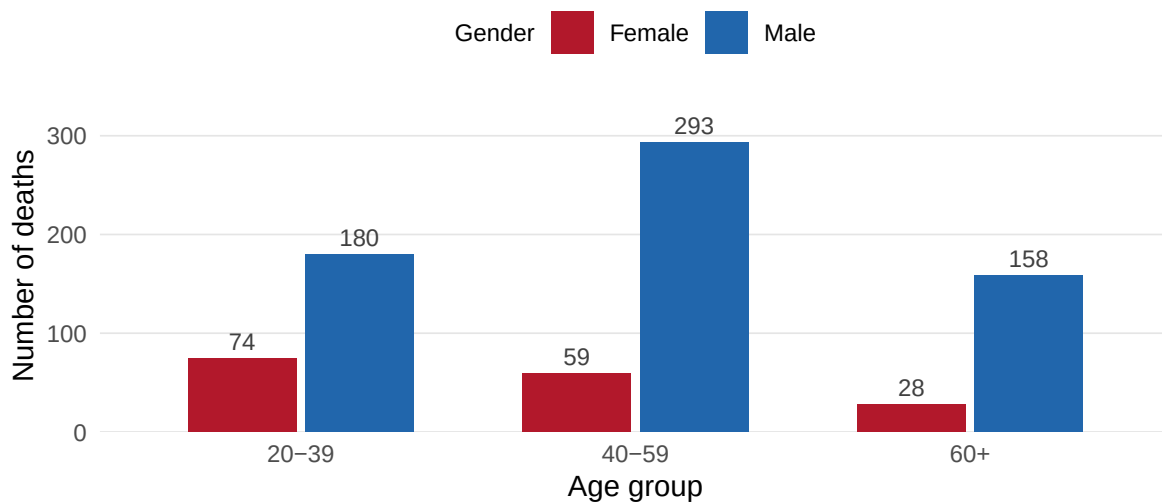


Figure 3: Deaths of people experiencing homelessness in Toronto by age group and gender, 2022 to 2024. The under-20 group is omitted because every cell for that age group is suppressed, and deaths with an unknown or unrecorded age group are also omitted. Deaths recorded with an unknown gender are also omitted from this figure.

2.6 Cause of death and age

Figure 4 shows the distribution of the causes of death across the age groups. The shape of the distribution changes across the panels. Among people aged 20 to 39, acute drug toxicity accounts for 198 of 256 deaths, or 77%, while the next largest cause, unknown, accounts for 14. Among people aged 40 to 59 it accounts for 210 of 352, or 60%. Among people aged 60 and over the count falls to 45 of 186, or 24%, and drug toxicity is only narrowly the largest single category, ahead of cardiovascular disease at 43 and deaths of unknown cause at 42. Taking cardiovascular disease and other diseases together, disease accounts for 78 deaths in that age group against 45 from drug toxicity.

Two further features are worth drawing out. Deaths of unknown cause rise with age, from 14 among people aged 20 to 39 to 42 among people aged 60 and over, and they make up 19 of the 31 deaths where age itself was not recorded, so the least documented deaths are concentrated among the oldest and the least identified.

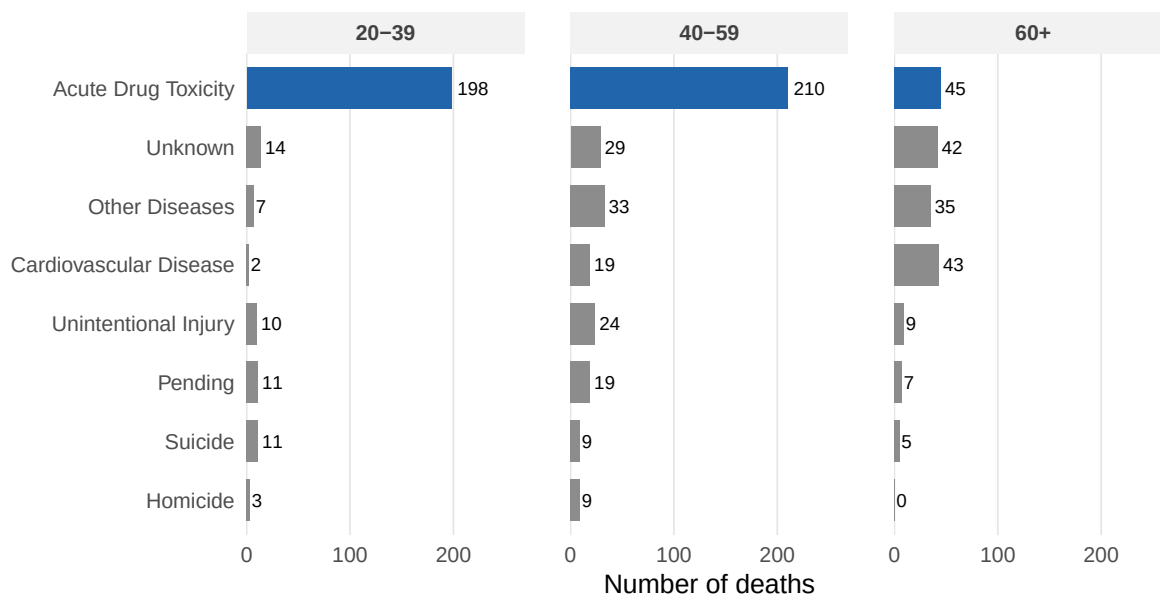


Figure 4: Deaths of people experiencing homelessness in Toronto by cause of death and age group, 2022 to 2024. The under-20 group is omitted because every cell for that age group is suppressed; deaths with an unknown or unrecorded age group are also omitted.

The published records describe two different patterns of death. Before the age of 60, deaths of people experiencing homelessness in Toronto are dominated by acute drug toxicity. From 60 onward, cardiovascular and other diseases together account for more deaths than drug toxicity does. Men account for most deaths in every age group. These are counts of deaths known to Toronto Public Health rather than all deaths, they carry no denominator, and they say nothing about people under 20, whose deaths are recorded but withheld from the published file.

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A Additional data details

A.1 Cleaning

The raw resource records Count as text rather than as a number, because cells recording individuals aged less than 20 contain the string “Suppressed” in place of a value. Cleaning identifies which cells were suppressed and replaces their Count with a missing value, so that later summaries can exclude them by filtering on missingness rather than on the string itself.

is_suppressed is derived from the original Count column before it is modified, so it reflects what the portal actually published rather than being inferred afterward from a missing value — this distinguishes a cell the portal suppressed from one that might be missing for some other reason.

The Final_Age (group) column was renamed to age_group in the cleaning script for convenience.

A.2 Tests

The simulated data are tested in Python, using polars and numpy, against the same distributional parameters used to generate them. Following Alexander (2023), writing these tests before acquiring the real data forces the structure of the dataset — its columns, its categories, and the range each one can take — to be specified in advance rather than discovered after the fact. Three checks are applied: that the simulated data frame has the expected columns and that every column takes only values from its expected set; that the observed proportion of each category falls within a tolerance of its simulated probability, using the binomial standard

error at each proportion rather than a single fixed tolerance across categories of very different sizes; and that the total count, and the count for each year, match the number of deaths the simulation was told to generate.

The real data are also tested in Python, using polars and numpy. The first confirms that both the raw and cleaned data have the columns the rest of the pipeline assumes, and that each column has the type it should: Year of death and `_id` as integers, Subgroup, age group, and Gender as text, and, in the cleaned data, `is_suppressed` as a boolean and Count as a number rather than the text it is in the raw file, since the raw resource records suppressed cells as the string “Suppressed” rather than leaving them blank. The second confirms that every value the dataset’s documentation describes actually appears at least once in the cleaned data — not only that no unexpected value has crept in, but that no described category, such as a rarer cause of death, has silently disappeared during cleaning. The third confirms that a cell is flagged `is_suppressed` exactly when, and only when, its Count is missing. This is checked in both directions within the cleaned data, and cross-checked against the raw file: the number of cells the portal marked “Suppressed” must equal the number flagged `is_suppressed` after cleaning, so a bug that mislabelled or dropped a suppression cannot pass silently. The fourth confirms that cleaning did not alter the data itself. The raw and cleaned datasets must have the same number of rows, and the total number of deaths must agree between them once suppressed cells, whose counts are unknown, are excluded from both sides.